

Chulhong Min

PRINCIPAL RESEARCH SCIENTIST AND TECH LEAD · NOKIA BELL LABS, CAMBRIDGE, UK

Broers Building, 21 J J Thomson Avenue, Cambridge, CB3 0FA, United Kingdom

✉ chulhong.min@gmail.com | 🏠 chulhongmin.com | 🎓 Google Scholar | 🌐 chulhong-min

Professional Summary

Principal Research Scientist and Tech Lead with extensive experience developing and deploying ML- and LLM-driven systems across heterogeneous compute platforms, including CPUs, GPUs, and NPUs. I specialize in model efficiency, on-device inference, and resource orchestration for distributed and multimodal systems across diverse compute and energy constraints.

I have a proven track record of translating advanced research into deployable solutions, working closely with cross-functional teams to deliver scalable, low-power, and reliable AI systems. My work bridges foundational research and real-world deployment, enabling AI solutions that support human-centred intelligence.

Professional Experience

Principal Research Scientist & Tech Lead

Nokia Bell Labs, Cambridge, UK

JUL. 2022 - PRESENT

- Led research and development of distributed AI systems for multi-modal sensing across wearables, edge, and IoT devices.
- Designed and evaluated adaptive ML and LLM-driven inference pipelines for heterogeneous edge-cloud environments, with a focus on accuracy, latency, reliability, and resource efficiency.
- Translated research prototypes into user-facing AI systems through close collaboration with cross-functional teams, including real-world deployments at scale (e.g., European Athletics U23 Championships).
- Built core AI infrastructure from scratch, including multi-tenant model execution runtimes, resource-aware schedulers, and heterogeneous accelerator (CPU/GPU/NPU) orchestration layers.
- Mentored junior researchers, shaping research direction, technical standards, and team processes.

Research Scientist / Member of Technical Staff

Nokia Bell Labs, Cambridge, UK

MAR. 2017 - JUN. 2022

- Conducted applied research on edge and on-device AI systems, developing frameworks for adaptive execution, distributed inference, and multi-tenant ML workloads under resource constraints.
- Co-led the eSense project, pioneering the first open earable sensing platform adopted by 70+ research labs worldwide and establishing early leadership in earable and wearable AI research.
- Built academic-industrial collaborations, including joint research programs with University of Cambridge (announcement).

Visiting Researcher

University of Cambridge, UK

MAR. 2019 - FEB. 2021

Postdoctoral Researcher

KAIST, Daejeon, South Korea

MAR. 2016 - DEC. 2016

- Conducted foundational research on mobile, wearable, and embedded AI systems, focusing on continuous human sensing, system optimization, and resource-aware ML execution.

Education

KAIST (Korea Advanced Institute of Science and Technology)

Daejeon, South Korea

PH.D. IN COMPUTER SCIENCE

Feb. 2016

- Advisor: Dr. June-hwa Song
- [Outstanding Ph.D Thesis Award](#), School of Computing

KAIST (Korea Advanced Institute of Science and Technology)

Daejeon, South Korea

M.S. IN COMPUTER SCIENCE

Aug. 2009

KAIST (Korea Advanced Institute of Science and Technology)

Daejeon, South Korea

B.S. IN COMPUTER SCIENCE

Aug. 2007

Research Experience

Paper tags (e.g., [J.17]) refer to entries in the Appendix, while links to other items are provided inline.

Multi-modal Sensory LLM Frameworks for Edge/Cloud Collaborative Deployment

Nokia Bell Labs
2025–Present

- This research investigates how Large Language Models (LLMs) can be extended to multi-modal sensory data from wearables under compute, memory, and energy constraints. We design a modular edge–cloud framework that characterizes resource demands, enables adaptive partitioning, and supports scalable distributed inference across heterogeneous platforms.

Ultra-efficient Runtime for On-device AI Acceleration

Nokia Bell Labs
2023–Present

BLUEPRINT, BLOG, PAPERS ([J.17], [C.22], [W.17], [W.15], [W.14], [W.12])

- This project develops an AI-native runtime for ultra-low-power wearables equipped with tiny AI accelerators. The research advances heterogeneous accelerator orchestration, memory virtualization, and software-defined thermal management, enabling efficient multi-model inference and continuous AI services on screenless wearables

On-the-fly Device-to-Device Authentication for Wearable AI Systems

Nokia Bell Labs
2023–Present

PAPERS ([C.23], [J.15], [W.16])

- This work explores secure and adaptive authentication mechanisms for tiny, screenless wearables that lack biometric sensors. We design on-the-fly, context-aware authentication strategies that enable dynamic device collaboration while balancing security, usability, and energy efficiency.

Distributed Inference and Deployment Framework for Edge Devices

Nokia Bell Labs
2021–2024

PAPERS ([J.16], [J.14], [J.13], [J.11], [J.10], [C.18], [C.16], [W.07])

- This research introduces a multi-tenant runtime for adaptive model execution and distributed inference on edge devices. The framework integrates MLOps, resource-aware scheduling, and redundancy-aware inference, demonstrating how collaborative edge systems can deliver scalable and reliable AI analytics.

Camera-as-a-Service: Real-world Validation of Edge AI

Nokia Bell Labs
2022–2023

PROJECT, VIDEO, NOKIA CEO'S MESSAGE, NOKIA BELL LABS PRESIDENT'S BLOG

- This deployment validates edge AI research in real-world settings. Our edge AI system was evaluated at large-scale live events, *European Athletics U23 Championships*, demonstrating robustness, scalability, and end-to-end reliability of collaborative video analytics pipelines under operational constraints.

eSense - Open Earable Platform for Human Sensing

Nokia Bell Labs
2017–2019

PROJECT, PAPERS ([J.09], [C.14], [C.13], [J.05], [W.05], [W.04], [D.10], [D.09], [D.08], [P.03])

- This project introduced the world's first open earable sensing platform, integrating audio, motion, and BLE signals for behavior analysis and well-being applications. Over 1,000 devices were distributed to 70+ research labs worldwide, catalyzing early research in earable AI.

Mobile, Wearable, and Embedded AI Systems

KAIST

PAPERS ([J.08], [C.10], [J.04], [J.03], [C.09], [C.08], [C.07], [J.02], [C.05], [C.04], [C.03], [J.01], [C.02], [C.01], [T.02], [T.01], [D.06], [D.05], [D.04], [P.02], [D.03], [P.01], [D.02], [F.01], [I.02], [D.01], [I.01])

- During my PhD, I developed foundational systems for continuous human sensing on mobile, wearable, and embedded platforms. This work addressed challenges in resource-aware execution, device collaboration, and sensing reliability, shaping early research on life-immersive AI systems.

Mobile, Social, and Earable Applications

Nokia Bell Labs, KAIST

PAPERS ([C.21], [C.20], [J.07], [C.15], [C.11], [C.06], [W.13], [W.10], [W.09], [W.06], [W.02], [W.01], [P.07], [P.06], [P.05], [P.05], [D.07], [V.01])

2007–Present

- As a systems researcher, I enjoyed developing innovative mobile, social, and earable applications on top of the systems I built. This hands-on work inspired new design insights and revealed novel requirements for next generation systems.

Datasets

Nokia Bell Labs, KAIST

SMARTWATCH [C.07], AMBIENT ACOUSTIC [C.17], FATIGUESet [C.19], WEEE [J.13]

2007–Present

- Released multiple publicly available datasets to support reproducible research in wearable sensing, multi-modal learning, and human-centered AI systems.

Publications & Intellectual Property

- 50+ peer-reviewed publications (2,000+ citations) in leading AI and systems venues, including NeurIPS, SenSys, MobiSys, UbiComp, and IEEE TMC.
- 20+ granted or pending patents translating research outcomes into deployable technologies for on-device AI, ML model optimization, and cross-device collaboration.
- The full list of publications and patents is provided in the Appendix.

SELECTIVE PUBLICATIONS

Taesik Gong, SiYoung Jang, Utku Günay Acer, Fahim Kawsar, **Chulhong Min**, *Synergy: Towards On-Body AI via Tiny AI Accelerator Collaboration on Wearables*, In IEEE Transactions on Mobile Computing (TMC), Volume 24, Issue: 10, pp 9319-9333.

Taesik Gong, Fahim Kawsar, **Chulhong Min**, *DEX: Data Channel Extension for Efficient CNN Inference on Tiny AI Accelerators*, In Proc. Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS 2024), Vancouver, Canada, December 2024.

Juheon Yi, Utku Gunay Acer, Fahim Kawsar, **Chulhong Min**, *Argus: Enabling Cross-Camera Collaboration for Video Analytics on Distributed Smart Cameras*, In IEEE Transactions on Mobile Computing (TMC), Volume 24, Issue: 1, pp 117-134.

Chulhong Min, Akhil Mathur, Alessandro Montanari, Fahim Kawsar, *SensiX: A System for Best-effort Inference of Machine Learning Models in Multi-device Environments*, In IEEE Transactions on Mobile Computing (TMC), Volume 22, Issue 9, pp 5525-5538.

Yash Jain, Chi Ian Tang, **Chulhong Min**, Fahim Kawsar, Akhil Mathur, *ColloSSL: Collaborative Self-Supervised Learning for Human Activity Recognition*, In Proc. ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (ACM IMWUT). Volume 6, Issue 1, Article 17 (March 2022).

Chulhong Min, Alessandro Montanari, Akhil Mathur, Fahim Kawsar, *A Closer Look at Quality-Aware Runtime Assessment of Sensing Models in Multi-Device Environments*, In Proc. the 17th ACM Conference on Embedded Networked Sensor Systems (ACM SenSys 2019), New York, USA, Nov. 10-13, 2019.

Honors & Awards

2024	Best Paper Award , ACM HumanSys	Hangzhou, China
2022	Best Paper Award , ACM AIChallengeloT	Boston, USA
2021	Best Student Paper Award , ICMU	Tokyo, Japan
2019	Honorable Mention , ACM MobileHCI	Taipei, Taiwan
2016	Outstanding Ph.D. Thesis Award , KAIST, School of Computing	Daejeon, South Korea
2014	Best Paper Award , ACM CSCW	Baltimore, MD, USA
2013	Qualcomm Fellowship , Qualcomm	Daejeon, South Korea
2013	Best Demo Award , ACM HotMobile	Georgia, USA

Academic & Community Leadership

Leadership roles in top-tier systems and AI research communities, spanning conference organization, editorial service, and program committees.

- **Vice General Chair**, ACM MobiSys 2026 (Cambridge, UK)
- **Editorial Board Member**, IEEE Pervasive Computing
- **Associate Editor**, ACM IMWUT
- **Program Committee Member** for multiple conferences, with sustained service across multiple years at ACM MobiSys, ACM MobiCom, and ACM SenSys

Technical Skills

AI & Machine Learning	LLM adaptation, multi-modal learning, efficiency-aware model design , Tensorflow, PyTorch
Edge & On-device AI	Quantisation, pruning, accelerator-aware optimisation , CPU/GPU/NPU/TPU
Distributed & Systems ML	Distributed inference, parallel execution, multi-tenant scheduling
Programming	Python, C/C++, Bash , Linux, Git
Research Engineering	Dataset design, experimental evaluation, reproducibility , IEEE/ACM publications, patent filing

APPENDIX

Publications

2025

- [J.18] Joseph A. Paradiso, Oliver Amft, **Chulhong Min**, *Defining a New Cross Reality*, IEEE Pervasive Computing, Volume: 24, Issue: 4, pp. 7–9.
- [J.17] Taesik Gong, SiYoung Jang, Utku Günay Acer, Fahim Kawsar, **Chulhong Min**, *Synergy: Towards On-Body AI via Tiny AI Accelerator Collaboration on Wearables*, IEEE Transactions on Mobile Computing (TMC), Volume: 24, Issue: 10, pp. 9319–9333.
- [C.23] Adiba Orzikulova, Diana A. Vasile, Chi Ian Tang, Fahim Kawsar, Sung-ju Lee, **Chulhong Min**, *BioQ: Towards Context-Aware Multi-Device Collaboration with Bio-cues*, ACM Conference on Embedded Networked Sensor Systems (ACM SenSys 2025), Irvine, USA, May 2025.
- [W.17] Changmin Jeon, Taesik Gong, Juheon Yi, Fahim Kawsar, **Chulhong Min**, *TinyMem: Boosting Multi-DNN Inference on Tiny AI Accelerators with Weight Memory Virtualization*, International Workshop on Mobile Computing Systems and Applications (ACM HotMobile 2025), Palm Springs, California, USA, Feb 2025.
- [W.16] Sujin Han, Diana A. Vasile, Fahim Kawsar, **Chulhong Min**, *SecuWear: Secure Data Sharing Between Wearable Devices*, Workshop on Security and Privacy in Standardized IoT (SDIoTSec 2025) in conjunction with NDSS 2025, San Diego, California, Feb 2025.

2024

- [C.22] Taesik Gong, Fahim Kawsar, **Chulhong Min**, *DEX: Data Channel Extension for Efficient CNN Inference on Tiny AI Accelerators*, Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS 2024), Vancouver, Canada, Dec 2024.
- [W.15] Yushan Huang, Taesik Gong, SiYoung Jang, Fahim Kawsar, **Chulhong Min**, *Energy Characterization of Tiny AI Accelerator-Equipped Microcontrollers*, International Workshop on Human-centered Sensing, Networking, and Multi-Device Systems (HumanSys 2024) in conjunction with ACM SenSys 2024, Hangzhou, China, Nov 2024. [\[Best paper award\]](#)
- [J.16] Juheon Yi, Utku Günay Acer, Fahim Kawsar, **Chulhong Min**, *Argus: Enabling Cross-Camera Collaboration for Video Analytics on Distributed Smart Cameras*, IEEE Transactions on Mobile Computing (TMC), Volume: 24, Issue: 1, pp. 117–134.
- [J.15] Diana A. Vasile, Fahim Kawsar, **Chulhong Min**, *Emerging Paradigms in Wearable Security: Adaptable and Secure Sandboxing for On-the-Fly Collaboration Among Wearables*, IEEE Security & Privacy. Volume: 22, Issue: 6, pp. 30–38.
- [W.14] SiYoung Jang, Fahim Kawsar, **Chulhong Min**, *Thermal Characterization of AI Applications on AI Accelerators-equipped Microcontrollers*, International Workshop on Body-Centric Computing Systems (BodySys 2024) in conjunction with ACM MobiSys 2024, Tokyo, Japan, Jun 2024.
- [A.03] **Chulhong Min**, Utku Günay Acer, SiYoung Jang, Sangwon Choi, Diana A. Vasile, Taesik Gong, Juheon Yi, Fahim Kawsar, *An AI-Native Runtime for Multi-Wearable Environments*, arXiv preprint arXiv:2403.17863 (2024)
- [A.02] Adiba Orzikulova, Diana A. Vasile, Fahim Kawsar, **Chulhong Min**, *Time-bound Contextual Bio-ID Generation for Minimalist Wearables*, arXiv preprint arXiv:2403.00889 (2024)

2023

- [C.21] Euihyeok Lee, **Chulhong Min**, Jaeseung Lee, Jin Yu, Seungwoo Kang, *GrooveMeter: Enabling Music Engagement-aware Apps by Detecting Reactions to Daily Music Listening via Earable Sensing*, ACM International Conference on Multimedia (MM), Ottawa, Canada, Oct 2023.
- [J.14] **Chulhong Min**, Akhil Mathur, Utku Günay Acer, Alessandro Montanari, Fahim Kawsar, *SensiX++: Bringing MLOps and Multi-tenant Model Serving to Sensory Edge Devices*, ACM Transactions on Embedded Computing Systems (TECS), Volume: 22, Issue 6, Article No.: 98, pp. 1–27.
- [J.13] **Chulhong Min**, Akhil Mathur, Alessandro Montanari, Fahim Kawsar, *SensiX: A System for Best-effort Inference of Machine Learning Models in Multi-device Environments*, IEEE Transactions on Mobile Computing (TMC), Volume: 22, Issue 9, pp 5525–5538.
- [W.13] Bhawana Chhagani, Utku Günay Acer, SiYoung Jang, Fahim Kawsar, **Chulhong Min**, *Cocoon: On-body Microphone Collaboration for Spatial Awareness*, International Workshop on Mobile Computing Systems and Applications (ACM HotMobile 2023), Orange County, California, USA, Feb 2023.

2022

- [W.12] Arthur Moss, Hyunjong Lee, Lei Xun, **Chulhong Min**, Fahim Kawsar, Alessandro Montanari, *Ultra-low Power DNN Accelerators for IoT: Resource Characterisation of the MAX78000*, ACM Workshop on Challenges in Artificial Intelligence and Machine Learning for Internet of Things (AIChallengeIoT 2022) in conjunction with ACM SenSys 2022, Boston, USA, Nov 2022. [Best paper award]
- [J.12] Shkurta Gashi, **Chulhong Min**, Alessandro Montanari, Silvia Santini, Fahim Kawsar, *A Multi-device and Multi-modal Dataset for Human Energy Expenditure Estimation using Wearable Devices*, Nature Scientific Data 9, Article No.: 537 (2022).
- [J.11] Utku Günay Acer, Marc Van Den Broeck, **Chulhong Min**, Mallesham Dasari, Fahim Kawsar, *The City as a Personal Assistant: Turning Urban Landmarks into Conversational Agents for Serving Hyper Local Information*, ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (ACM IMWUT). Volume: 6, Issue 2, Article No.: 40, pp. 1–21.
- [J.10] Yash Jain, Chi Ian Tang, **Chulhong Min**, Fahim Kawsar, Akhil Mathur, *ColloSSL: Collaborative Self-Supervised Learning for Human Activity Recognition*, ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (ACM IMWUT). Volume: 6, Issue 1, Article No.: 17, pp. 1–28.
- [C.20] Jing Yang, **Chulhong Min**, Akhil Mathur, Fahim Kawsar, *SleepGAN: Towards Personalized Sleep Therapy Music*, IEEE International Conference on Acoustics, Speech and Signal Processing (IEEE ICASSP), Singapore, May 2022.
- [J.09] Andrea Ferlini, Alessandro Montanari, **Chulhong Min**, Hongwei Li, Ugo Sassi, Fahim Kawsar, *In-Ear PPG for Vital Signs*, IEEE Pervasive Computing, Volume: 21, Issue 1, pp. 65–74.

2021

- [C.19] Manasa Kalanadhabhatta, **Chulhong Min**, Alessandro Montanari, Fahim Kawsar, *FatigueSet: A Multi-modal Dataset for Modeling Mental Fatigue and Fatigability*, EAI International Conference on Pervasive Computing Technology for Healthcare (PervasiveHealth), Dec 2021.
- [C.18] SiYoung Jang, Utku Günay Acer, **Chulhong Min**, Fahim Kawsar, *Deploying Collaborative Machine Learning Systems in Edge with Multiple Cameras*, 13th International Conference on Mobile Computing and Ubiquitous Networking (ICMU), Nov 2021. [Best student paper award]
- [W.11] Juheon Yi, **Chulhong Min**, Fahim Kawsar, *Vision Paper: Towards Software-Defined Video Analytics with Cross-Camera Collaboration*, ACM Workshop on Challenges in Artificial Intelligence and Machine Learning for Internet of Things (AIChallengeIoT) in conjunction with ACM SenSys 2021, Coimbra, Portugal, Nov 2021.
- [W.10] Dongwoo Kim, **Chulhong Min**, Seungwoo Kang, *Towards Automatic Recognition of Perceived Level of Understanding on Online Lectures using Earables*, ACM Workshop on Earable Computing (ACM EarComp) in conjunction with ACM UbiComp 2021, Virtual, Sep 2021.
- [W.09] Matija Franklin, David Lagnado, **Chulhong Min**, Akhil Mathur, Fahim Kawsar, *Designing Memory Aids for Dementia Patients using Earables*, ACM Workshop on Earable Computing (ACM EarComp) in conjunction with ACM UbiComp 2021, Virtual, Sep 2021.
- [A.01] **Chulhong Min**, Akhil Mathur, Utku Günay Acer, Alessandro Montanari, Fahim Kawsar *SensiX++: Bringing MLOPs and Multi-tenant Model Serving to Sensory Edge Devices*, arXiv preprint arXiv:2109.03947 (2019)
- [W.08] Yash Jain, Chi Ian Tang, **Chulhong Min**, Fahim Kawsar, Akhil Mathur, *Group Supervised Learning: Extending Self-Supervised Learning to Multi-Device Settings*, ICML 2021 Workshop Self-Supervised Learning for Reasoning and Perception in conjunction with ICML 2021, Virtual, Jul 2021.

2020

- [P.07] Dongwoo Kim, **Chulhong Min**, Seungwoo Kang, *Poster Abstract: Towards Recognizing Perceived Level of Understanding for Online Lectures using Earables*, ACM Conference on Embedded Networked Sensor Systems (ACM SenSys), Yokohama, Japan, Nov 2020.
- [P.06] Euihyeok Lee, Dongwoo Kim, **Chulhong Min**, Seungwoo Kang, *Poster Abstract: Automatic Recognition of Vocal Reactions in Music Listening using Smart Earbuds*, ACM Conference on Embedded Networked Sensor Systems (ACM SenSys), Yokohama, Japan, Nov 2020.
- [C.17] Chunjong Park, **Chulhong Min**, Sourav Bhattacharya, Fahim Kawsar, *Augmenting Conversational Agents with Ambient Acoustic Contexts*, 22nd International Conference on Human-Computer Interaction with Mobile Devices and Services (MobileHCI), Virtual, Oct 2020.

[J.08] **Chulhong Min**, Youngki Lee, Chungkuk Yoo, Inseok Hwang, Younghyun Ju, Junehwa Song, Seungwoo Kang, *Scalable Power Impact Prediction of Mobile Sensing Applications at Pre-installation Time*, IEEE Transactions on Mobile Computing (TMC), Volume: 19, Issue 6, pp. 1448–1464.

2019

[C.16] **Chulhong Min**, Alessandro Montanari, Akhil Mathur, Fahim Kawsar, *A Closer Look at Quality-Aware Runtime Assessment of Sensing Models in Multi-Device Environments*, 17th ACM Conference on Embedded Networked Sensor Systems (ACM SenSys), New York, USA, Nov 2019.

[W.07] Mattia Antonini, Tran Huy Vu, **Chulhong Min**, Alessandro Montanari, Akhil Mathur, Fahim Kawsar, *Resource Characterisation of Personal-Scale Sensing Models on Edge Accelerators*, ACM Workshop on Challenges in Artificial Intelligence and Machine Learning for Internet of Things (AIChallengIoT) in conjunction with ACM SenSys 2019, New York, USA, Nov 2019.

[J.07] Inseok Hwang, Youngki Lee, Chungkuk Yoo, **Chulhong Min**, Dongsun Yim, John Kim, *Towards Interpersonal Assistants: Next-generation Conversational Agents*, IEEE Pervasive Computing, Volume: 18, Issue 2, pp. 21–31.

[C.15] **Chulhong Min**, Euihyeok Lee, Souneil Park, Seungwoo Kang, *Tiger: Wearable Glasses for the 20-20-20 Rule to Alleviate Computer Vision Syndrome*, 21st International Conference on Human-Computer Interaction with Mobile Devices and Services (MobileHCI), Taipei, Taiwan, Oct 2019. [\[Honorable Mention\]](#)

[C.14] **Chulhong Min**, Akhil Mathur, Alessandro Montanari, Fahim Kawsar, *An Early Characterisation of Wearing Variability on Motion Signals for Wearables*, International Symposium on Wearable Computers (ISWC), London, UK, Sep 2019.

[W.06] Chungkuk Yoo, Seungwoo Kang, Inseok Hwang, **Chulhong Min**, Seonghoon Kim, Wonjung Kim, Junehwa Song, *Mom, I see You Angry at Me! Designing a Mobile Service for Parent-child Conflicts by In-situ Emotional Empathy*, ACM Workshop on Mobile Systems for Computational Social Science (ACM MCSS) in conjunction with ACM MobiSys 2019, Seoul, South Korea, Jun 2019.

[C.13] Seungchul Lee, **Chulhong Min**, Alessandro Montanari, Akhil Mathur, Youngjae Chang, Junehwa Song, Fahim Kawsar, *Automatic Smile and Frown Recognition with Kinetic Earables*, 10th Augmented Human International Conference (AH), Reims Champagne-Ardenne, France, Mar 2019.

[J.06] Mohammed Alloulah, Anton Isopoussu, **Chulhong Min**, Fahim Kawsar, *On Tracking the Physicality of Wi-Fi: A Subspace Approach*, IEEE Access, Volume: 7, pp. 19965–19978.

2018

[C.12] **Chulhong Min**, Mohammed Alloulah, Fahim Kawsar, *An Early Resource Characterisation of Wi-Fi Sensing on Residential Gateways*, ACM International Conference on Energy-Efficient Built Environments (ACM BuildSys), Shenzhen, China, Nov 2018.

[D.10] Fahim Kawsar, **Chulhong Min**, Akhil Mathur, Alessandro Montanari, Utku Günay Acer, Marc Van den Broeck, *Demo Abstract: eSense - Open Earable Platform for Human Sensing*, ACM Conference on Embedded Networked Sensor Systems (ACM SenSys), Shenzhen, China, Nov 2018.

[J.05] Fahim Kawsar, **Chulhong Min**, Akhil Mathur, Alessandro Montanari, *Earables for Personal-Scale Behavior Analytics*, IEEE Pervasive Computing, Volume: 17, Issue: 3, pp. 83–89.

[W.05] **Chulhong Min**, Alessandro Montanari, Akhil Mathur, Seungchul Lee, Fahim Kawsar, *Cross-Modal Approach for Conversational Well-being Monitoring with Multi-Sensory Earables*, ACM Workshop on Computing for Well-being (ACM Well-Comp) in conjunction with ACM UbiComp 2018, Singapore, Singapore, Oct 2018.

[D.09] Fahim Kawsar, **Chulhong Min**, Akhil Mathur, Alessandro Montanari, Utku Günay Acer, Marc Van den Broeck, *Demo: eSense - Open Earable Platform for Human Sensing*, ACM International Joint Conference and International Symposium on Pervasive and Ubiquitous Computing and Wearable Computers (ACM UbiComp), Singapore, Singapore, Oct 2018.

[P.05] Euihyeok Lee, **Chulhong Min**, Seungwoo Kang, *Poster: Towards a Wearable Assistant to Prevent Computer Vision Syndrome*, ACM International Joint Conference and International Symposium on Pervasive and Ubiquitous Computing and Wearable Computers (ACM UbiComp), Singapore, Singapore, Oct 2018.

[P.04] Seungchul Lee, Saumay Pushp, **Chulhong Min**, Junehwa Song, *Poster: Exploring Relationship-aware Dynamic Message Screening for Mobile Messengers*, ACM International Joint Conference and International Symposium on Pervasive and Ubiquitous Computing and Wearable Computers (ACM UbiComp), Singapore, Singapore, Oct 2018.

[W.04] **Chulhong Min**, Akhil Mathur, Fahim Kawsar, *Exploring Audio and Kinetic Sensing on Earable Devices*, ACM Workshop on Wearable Systems and Applications (ACM WearSys) in conjunction with ACM MobiSys 2018, Munich, Germany, Jun 2018.

[D.08] Fahim Kawsar, **Chulhong Min**, Akhil Mathur, Marc Van den Broeck, Utku Günay Acer, Claudio Forlivesi, *Demo: eSense - Open Earable Platform for Human Sensing*, ACM Annual International Conference on Mobile Systems, Applications, and Services (ACM MobiSys), Munich, Germany, Jun 2018.

[P.03] **Chulhong Min**, Akhil Mathur, Fahim Kawsar, *Poster: Audio-Kinetic Model for Automatic Dietary Monitoring with Earable Devices*, ACM Annual International Conference on Mobile Systems, Applications, and Services (ACM MobiSys), Munich, Germany, Jun 2018.

2017

[C.11] Bumsoo Kang, **Chulhong Min**, Wonjung Kim, Inseok Hwang, Chunjong Park, Seungchul Lee, Sung-ju Lee, Junehwa Song, *Zaturi: We Put Together the 25th Hour for You. Create a Book for Your Baby*, ACM Conference on Computer-Supported Cooperative Work and Social Computing (ACM CSCW), Portland, Oregon, USA, Feb 2017.

[D.07] Bumsoo Kang, Wonjung Kim, Inseok Hwang, Chunjong Park, Seungchul Lee, **Chulhong Min**, Sung-Ju Lee, Junehwa Song, *Zaturi: Blending Hours Spent at Work and Hours Devoted to Children*, ACM Conference on Computer-Supported Cooperative Work and Social Computing (ACM CSCW), Portland, Oregon, USA, Feb 2017.

2016

[C.10] **Chulhong Min**, Seungchul Lee, Changhun Lee, Youngki Lee, Seungwoo Kang, Seungpyo Choi, Wonjung Kim, Junehwa Song, *PADA: Power-aware Development Assistant for Mobile Sensing Applications*, ACM International Joint Conference on Pervasive and Ubiquitous Computing (ACM UbiComp), Heidelberg, Germany, Sep 2016.

[J.04] Youngki Lee, Seungwoo Kang, **Chulhong Min**, Younghyun Ju, Inseok Hwang, Junehwa Song, *CoMon+: A Cooperative Context Monitoring System for Multi-Device Personal Sensing Environments*, IEEE Transactions on Mobile Computing (TMC), Volume: 15, Issue: 8, pp. 1908–1924.

[J.03] **Chulhong Min**, Youngki Lee, Chungkuk Yoo, Seungwoo Kang, Inseok Hwang, Junehwa Song, *PowerForecaster: Predicting Power Impact of Mobile Sensing Applications at Pre-installation Time*, GetMobile: Mobile Computing and Communication, Volume: 20, Issue: 1, pp. 30–33.

[T.01] **Chulhong Min**, *User Support for Battery Management of Continuous Sensing Applications*, Ph.D. Thesis, School of Computing, KAIST, Feb 2016. [\[Outstanding Ph.D. Thesis Award\]](#)

2015

[C.09] **Chulhong Min**, Youngki Lee, Chungkuk Yoo, Seungwoo Kang, Sangwon Choi, Pillsoon Park, Inseok Hwang, Younghyun Ju, Seungpyo Choi, Junehwa Song, *PowerForecaster: Predicting Smartphone Power Impact of Continuous Sensing Applications at Pre-installation Time*, ACM Conference on Embedded Networked Sensor Systems (ACM SenSys), Seoul, Korea, Nov 2015.

[D.06] **Chulhong Min**, Chungkuk Yoo, Sangwon Choi, Pillsoon Park, Seungchul Lee, Changhun Lee, Seungpyo Choi, Seungwoo Kang, Youngki Lee, Inseok Hwang, Younghyun Ju, Junehwa Song, *Demo: User Support for Power Management of Continuous Sensing Applications*, ACM Conference on Embedded Networked Sensor Systems (ACM SenSys), Seoul, Korea, Nov 2015.

[C.08] **Chulhong Min**, Chungkuk Yoo, Inseok Hwang, Seungwoo Kang, Youngki Lee, Seungchul Lee, Pillsoon Park, Changhun Lee, Seungpyo Choi, Junehwa Song, *Sandra Helps You Learn: the More You Walk, the More Battery Your Phone Drains*, ACM International Joint Conference on Pervasive and Ubiquitous Computing (ACM UbiComp), Osaka, Japan, Sep 2015.

[C.07] **Chulhong Min**, Seungwoo Kang, Chungkuk Yoo, Jeehoon Cha, Sangwon Choi, Younghyun Oh, Junehwa Song, *Exploring Current Practices for Battery Use and Management of Smartwatches*, Proceedings of the 19th International Symposium on Wearable Computers (ISWC), Osaka, Japan, Sep 2015.

[W.03] Sebastian Deterding, Andrés Lucero, Jussi Holopainen, **Chulhong Min**, Adrian Cheok, Annika Waern, Steffen Walz, *Embarrassing Interactions*, Embarrassing Interactions 2015, in conjunction with ACM CHI 2015, Seoul, Korea, Apr 2015.

2014

[W.02] **Chulhong Min**, Saumay Pushp, Seungchul Lee, Inseok Hwang, Youngki Lee, Seungwoo Kang, Junehwa Song, *Uncovering Embarrassing Moments in In-situ Exposure of Incoming Mobile Messages*, ACM Workshop on Mobile Systems for Computational Social Science (ACM MCSS) in conjunction with ACM UbiComp 2014, Seattle, USA, Sep 2014.

[J.02] Youngki Lee, **Chulhong Min**, Younghyun Ju, Seungwoo Kang, Yunseok Rhee, Junehwa Song, *An Active Resource Orchestration Framework for PAN-Scale, Sensor-Rich Environments*, IEEE Transactions on Mobile Computing (TMC), Volume: 13, Issue: 3, pp. 596–610.

[C.06] Inseok Hwang, Chungkuk Yoo, Chanyou Hwang, Dongsun Yim, Youngki Lee, **Chulhong Min**, John Kim, Junehwa Song, *TalkBetter: family-driven mobile intervention care for children with language delay*, ACM conference on Computer supported cooperative work and social computing (ACM CSCW), Baltimore, USA, Feb 2014. [\[Best paper award\]](#)

2013

[W.01] SangJeong Lee, **Chulhong Min**, Chungkuk Yoo, Junehwa Song, *Understanding Customer Malling Behavior in an Urban Shopping Mall using Smartphones*, ACM Workshop on Mobile Systems for Computational Social Science (ACM MCSS) in conjunction with ACM UbiComp 2013, Zurich, Switzerland, Sep 2013.

[C.05] Youngki Lee, **Chulhong Min**, Chanyou Hwang, Jaeung Lee, Inseok Hwang, Younghyun Ju, Chungkuk Yoo, Miri Moon, Uichin Lee, Junehwa Song, *SocioPhone: Everyday Face-to-Face Interaction Monitoring Platform using Multi-phone Sensor Fusion*, ACM Annual International Conference on Mobile systems, applications, and services (ACM MobiSys), Taipei, Taiwan, Jun 2013.

[D.05] Youngki Lee, **Chulhong Min**, Chanyou Hwang, Jaeung Lee, Inseok Hwang, Younghyun Ju, Chungkuk Yoo, Miri Moon, Uichin Lee, Junehwa Song, *Demo - SocioPhone: Everyday Face-to-Face Interaction Monitoring Platform using Multi-phone Sensor Fusion*, ACM Annual International conference on Mobile systems, applications, and services (ACM MobiSys), Taipei, Taiwan, Jun 2013.

[D.04] **Chulhong Min**, Inseok Hwang, Jaeung Lee, Chanyou Hwang, Chungkuk Yoo, Miri Moon, Taiwoo Park, Changhoon Lee, Haechan Lee, Yuhwan Kim, Younghyun Ju, Youngki Lee, Uichin Lee, Junehwa Song, *ACM HotMobile 2013 Demo: Bringing In-situ Social Awareness to Mobile Systems: Conversational Turn Monitoring and its Applications*, MC2R '13: ACM Mobile Computing and Communication Review, Volume: 17, Number: 3 (2013) and ACM Workshop on Mobile Computing Systems and Applications (ACM HotMobile), Jekyll Island, USA, Feb 2013. [\[Best Demo Award\]](#)

2012

[C.04] Younghyun Ju, Youngki Lee, Jihyun Yu, **Chulhong Min**, Insik Shin, Junehwa Song, *SymPhoney: A Coordinated Sensing Flow Execution Engine for Concurrent Mobile Sensing Applications*, ACM Conference on Embedded Network Sensor Systems (ACM SenSys), Toronto, Canada, Nov 2012.

[P.02] Saumay Pushp, **Chulhong Min**, Youngki Lee, Chi Harold Lie, Junehwa Song, *Towards Crowd-aware Sensing Platform for Metropolitan Environments*, ACM Conference on Embedded Network Sensor Systems (ACM SenSys), Toronto, Canada, Nov 2012.

[C.03] Youngki Lee, Younghyun Ju, **Chulhong Min**, Seungwoo Kang, Inseok Hwang, Junehwa Song, *CoMon: Cooperative Ambience Monitoring Platform with Continuity and Benefit Awareness*, ACM International Conference on Mobile systems, applications, and services (ACM MobiSys), Low Wood Bay, Lake District, United Kingdom, Jun 2012.

[P.01] **Chulhong Min**, Wookhyun Han, Inseok Hwang, SangJeong Lee, Youngki Lee, Insik Shin, Junehwa Song, *Poster: Towards Mobile GPU-Accelerated Context Processing for Continuous Sensing Applications on Smartphones*, Annual International Conference on Mobile systems, applications, and services (ACM MobiSys), Low Wood Bay, Lake District, United Kingdom, Jun 2012.

[F.01] **Chulhong Min**, *SensorShader: Mobile GPU-Accelerated Context Processing Engine for Sensing Applications on Smartphones*, MobiSys Ph.D. Forum, Low Wood Bay, Lake District, United Kingdom, Jun 2012.

[D.03] Youngki Lee, Younghyun Ju, **Chulhong Min**, Jihyun Yu, Junehwa Song, *MobiCon: Mobile context monitoring platform, Incorporating Context-awareness to Smartphone-centric Personal Sensor Networks*, IEEE Communications Society Conference on Sensor, Mesh and Ad Hoc Communications and Networks (IEEE SECON), Seoul, Korea, Jun 2012.

[D.02] Youngki Lee, Younghyun Ju, **Chulhong Min**, Seungwoo Kang, Inseok Hwang, Junehwa Song, *Demo: SenseTogether- Cooperative Ambience Monitoring Platform with Continuity and Benefit Awareness*, Annual International Conference on Mobile systems, applications, and services (ACM MobiSys), Low Wood Bay, Lake District, United Kingdom, Jun 2012.

[J.01] Youngki Lee, Sitharam S. Iyengar, **Chulhong Min**, Younghyun Ju, Seungwoo Kang, Taiwoo Park, Jinwon Lee, Yunseok Rhee, Junehwa Song, *MobiCon: A Mobile Context-monitoring Platform*, Communications of the ACM (CACM), Volume: 55, Issue: 3, pp. 54–65.

[C.02] Younghyun Ju, **Chulhong Min**, Youngki Lee, Jihyun Yu, Junehwa Song, *An Efficient Dataflow Execution Method for Mobile Context Monitoring Applications*, IEEE International Conference on Pervasive Computing and Communications (PerCom), Lugano, Switzerland, Mar 2012.

2011

[I.02] **Chulhong Min**, Chungkuk Yoo, Youngki Lee, Junehwa Song, *Healthopia: Towards Your Well-Being in Everyday Life (invited paper)*, International Symposium on Applied Sciences in Biomedical and Communication Technologies (ISABEL 2011), Barcelona, Spain, Oct 2011.

[D.01] Youngki Lee, Younghyun Ju, **Chulhong Min**, Seungwoo Kang, Yunseok Rhee, Junehwa Song, *Demo: CoMon – Resource-aware Cooperative Context Monitoring System for Smartphone-centric Sensor-rich PANs*, ACM Annual International Conference on Mobile systems, applications, and services (ACM MobiSys), Washington, DC, USA, Jun 2011.

[I.01] Youngki Lee, **Chulhong Min**, Younghyun Ju, Saumay Pushp, Junehwa Song, *A Mobile Context Monitoring Platform for Pervasive Computing Environments (Invited paper)*, IEEE International Conference on Digital Ecosystems and Technologies (IEEE DEST), Daejeon, Korea, May 2011.

2010

[C.01] Seungwoo Kang, Youngki Lee, **Chulhong Min**, Younghyun Ju, Taiwoo Park, Jinwon Lee, Yunseok Rhee, Junehwa Song, *Orchestrator: An Active Resource Orchestration Framework for Mobile Context Monitoring in Sensor-rich Mobile Environments*, IEEE International Conference on Pervasive Computing and Communications (PerCom), Mannheim, Germany, Mar 2010.

Patents

GRANTED

Personalized models (GB2588689A)

Detection of facial expressions (US12299187B2, CN113557490B, WO2020182447A1, EP3709131A1 - pending)

System and method for automatic detection of music listening reactions, and mobile device performing the method (US20230171541A1, KR102735260B1)

Presence or activity detection (EP3459439)

Method and devices for processing sensor data by applying one or more processing pipelines to the sensor data (US11557898, WO2019091959A1)

Communication device for predicting power consumption of mobile application, communication system including same, method of predicting power consumption of mobile application and method of providing predicted power consumption of mobile application, using same (US10928877B2, WO2017155322A2, KR101758267B1)

Sensing in wireless communication systems (EP3624410)

Mobile apparatus, audio book creating system having the same and method of creating audio book using the same (KR102061619B1)

Language delay treatment system and control method for the same (US9875668B2)

Mobile device executing face-to-face interaction monitoring, method of monitoring face-to-face interaction using the same, and interaction monitoring system including the same, and mobile interaction monitoring application executed on the same (US9813879B2)

Mobile device for performing monitoring on cooperative situation, method of performing monitoring on cooperative situation using said mobile device, and system for performing monitoring on cooperative situation including said mobile device (US9756095B2, WO2014046369A1 - pending)

Mobile apparatus supporting cooperative context monitoring, method of cooperative context monitoring using the same and cooperative context monitoring system including the same (US9756095B2)

Mobile apparatus for executing sensing flow for mobile context monitoring, method of executing sensing flow using the same, method of context monitoring using the same and context monitoring system including the same (US9367664B2, KR101549002B1)

Mobile apparatus supporting context monitoring, method of monitoring context using the same and context monitoring system having the same (US8599710B2)

PENDING

Multi-camera image data processing (US20260025483A1)

Apparatus, method, and system for multi-device wearability-aware heat orchestration (US20250264933A1, CN120523674A, EP4607318A1, GB2638425A)

Method for multi-factor user authentication (US20250211586A1, EP4575849A1)

Machine learning model scheduler (US20250165295A1, EP4557096A1, GB2635567A)

Apparatus & method for multi-device authentication (US20250094559A1, EP4524786A1, GB2633624A)
Earphone Modes of Operation (US20240040299A1, EP4312436A1)
Object Identification (US20230394686A1, EP4290472A1)
Resource control (US20230269668A1, WO2022023618A1, EP4190042A4)
Resource assessment of sensors (US20220330896A1, CN114402179A, EP4031838A1, WO2021053444A1)
Model modification and deployment (US20210350280A1)

Talks

Invited talks and keynote presentations at leading universities, research institutes, and international conferences.

AI-powered Evolution in Collaborative Wearables with Multi-device, Multi-modal Systems, CS Department Special Seminar, POSTECH, Pohang, South Korea, Aug. 2024

AI-powered Evolution in Wearables with Multi-device, Multi-modal, Collaborative Systems, Keynote Speech, NetAISys, Tokyo, Japan, Jun. 2024

What's Next for AI Systems?, KAIST, Daejeon, South Korea, Dec. 2023

Building Collaborative AI Systems for Multi-device Environment, Guest Lecture, University of Southampton, Southampton, UK, Nov. 2022

Building AI Systems for Collaborative, Interactive, and Privacy-Preserving Applications, Special Seminar, POSTECH, Pohang, South Korea, Nov. 2022

Building Collaborative AI Systems for Multi-device Environments, CS Seminar, UNIST, Ulsan, South Korea, Nov. 2022

The Challenge and Future of Sensory AI Systems: Making On-device AI a Reality, Invited Talk, SIGMOBILE CEP, Online, Mar. 2022

Earable Computing for Personal-scale Behavioral Analytics, Invited Talk, Daejeon, South Korea, Jul. 2021

What is Next for Sensory AI Systems? A Journey toward Making AI Work in the Real World, Special Seminar, Yonsei University, Incheon, South Korea, Jun. 2021

What is Next for AI Systems? A Journey toward Making AI Work in the Real World, Special Seminar, POSTECH, Pohang, South Korea, May 2021

Sensory AI Software Platform for Multi-device Environments, Special Session Invited Talk, ICTC, Jeju Island, South Korea, Oct. 2020

Sensory AI Software Platform for Multi-device Environments, Industry Track Invited Talk, IEEE SMARTCOMP, Bologna, Italy, Sep. 2020

Making AI Work in Multi-device World, Guest Lecture, Seoul National University, Seoul, South Korea, Jun. 2019

Resource Orchestration Platform for Life-immersive Sensing Applications, Invited Talk, Future Internet Forum Workshop, Seoul, South Korea, Oct. 2016

Ecosystem-wide Support for Power Impact-Awareness for Mobile Sensing Applications, Invited Talk, Nokia Bell Labs, Antwerp, Belgium, Sep. 2016

Ecosystem-centric Power Management for Continuous Sensing Applications, Invited Talk, Naver Labs, Sungnam, South Korea, Feb. 2016

An Active Resource Use Orchestration Framework for Mobile Context Monitoring in Sensor-rich Mobile Environments, Invited Talk, Research Center for UX-oriented Software Platform, Daejeon, South Korea, Feb. 2013

CoMon: Cooperative Ambience Monitoring Platform with Continuity and Benefit Awareness, Invited Talk, POSTECH, Pohang, South Korea, Aug. 2012

EDITORSHIP

- 2023–Present **IEEE Pervasive Computing**, Editorial Board Member
2017–Present **ACM IMWUT**, Associate Editor

ORGANISING COMMITTEE

- | | | |
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| 2026 | ACM MobiSys , Vice General Chair | <i>Cambridge, UK</i> |
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| 2025 | ACM UbiComp/ISWC , Sponsorship and Industrial Relations Chairs | <i>Helsinki, Finland</i> |
| 2024 | ACM MobiSys , Student Travel Grants Chair | <i>Tokyo, Japan</i> |
| 2023 | ACM IASA , Sponsorship Chair | <i>San Antonio, USA</i> |
| 2022 | ACM UbiComp/ISWC , Workshop Chair | <i>Cambridge, UK</i> |
| 2021 | ACM EarComp , Program Chair | <i>Online</i> |
| 2020 | ACM SenSys , Poster and Demo Chair | <i>Yokohama, Japan</i> |
| 2019 | ACM EarComp , Local Arrangement | <i>London, UK</i> |
| 2019 | ACM WearSys , Program Chair | <i>Seoul, South Korea</i> |
| 2019 | ACM MobiSys , Demo and Video Chair | <i>Seoul, South Korea</i> |
| 2018 | ACM UbiComp , Publication Chair | <i>Singapore, Singapore</i> |
| 2017 | IEEE MDM , Local Organising Chair | <i>Daejeon, South Korea</i> |

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2021 **ACM HotMobile**, **COMSNETS**, **ICMU**, **MobiQuitous**, **MFSens**
2020 **ACM WearSys**, **WCNC**, **COMSNETS**
2019 **ACM MobiSys**, **ACM EarComp**, **PervasiveHealth**, **MUSICAL**
2017 **IEEE MDM**
2015 **Embarrassing Interactions**, **ACM MobiSys PhD Forum**